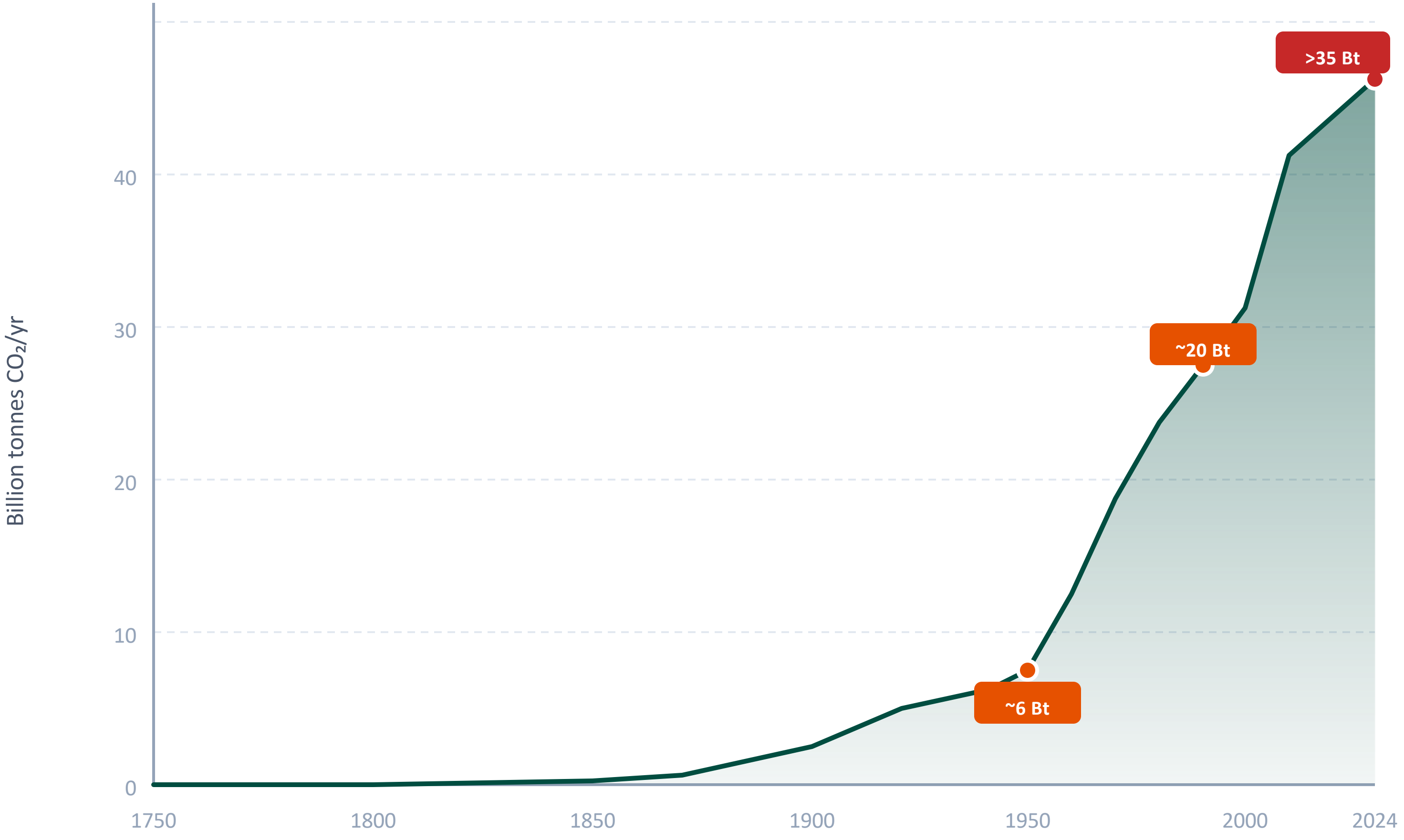
Global CO_2 Emissions: Who Emits, How Much, and What's Changing

Fossil Fuels & Industry, 1750–2024

Data: Global Carbon Budget 2025 via Our World in Data

Global CO_2 Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today

Key Insight: Global emissions have not yet peaked — growth has slowed but the absolute level continues to rise



Emission Trajectory

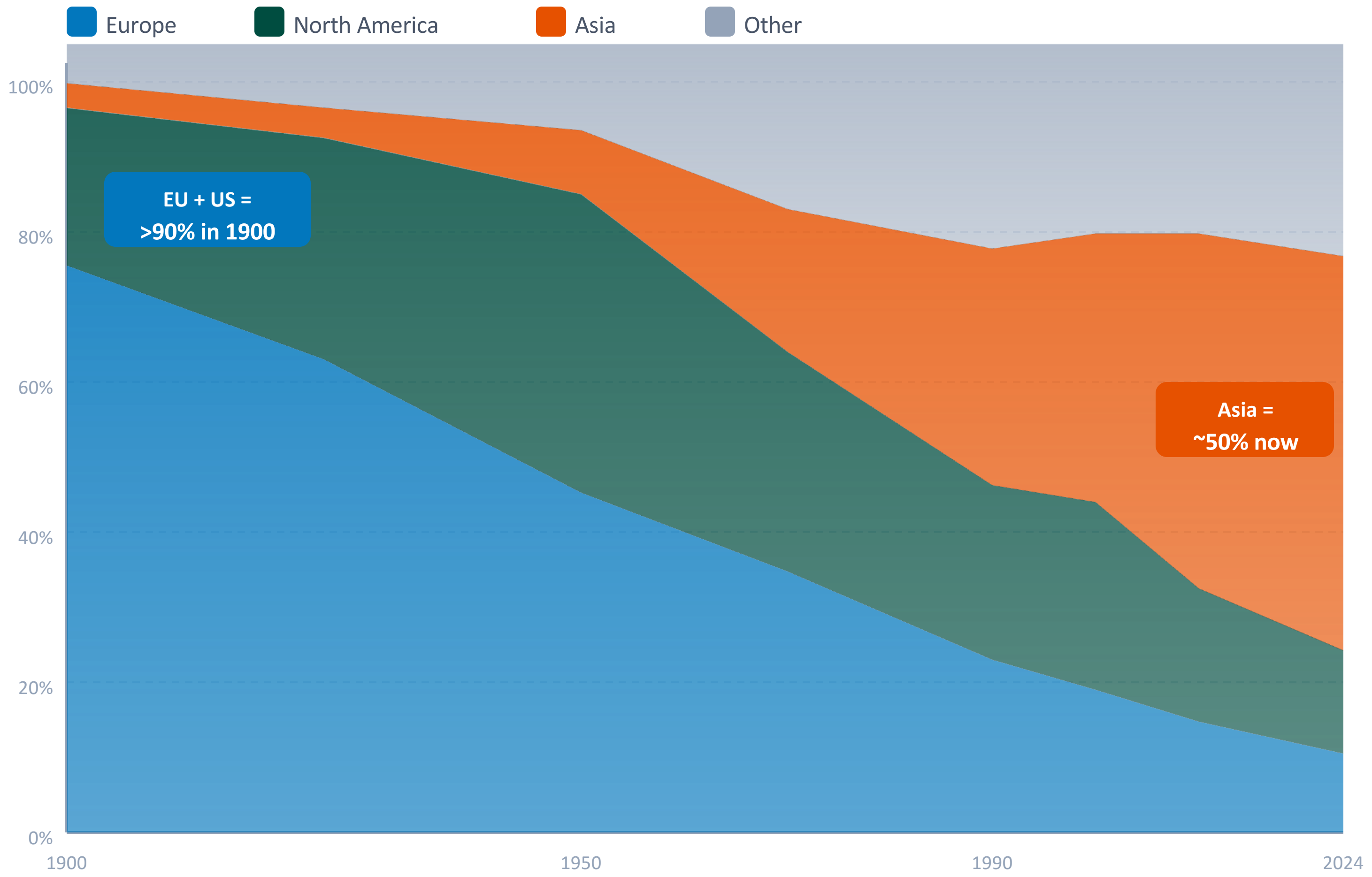
- Before industrialization: emissions were negligible
- 1950: ~6 Bt/year
- Post-WWII boom: rapid industrialization drives growth
- 1990: nearly quadrupled to **~20 Bt/year**
- 2024: exceeds 35 Bt/year
- Growth has slowed but **emissions have NOT peaked**

Land-use change emissions have declined slightly, but fossil fuels continue to dominate the total.

Source: Global Carbon Budget 2025 via Our World in Data

Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today

The center of gravity has shifted from West to East — Asia now produces roughly half of global emissions



The West's Declining Share

EU + US share in 1900

>90%

EU + US share today

<1/3

Asia's Rise

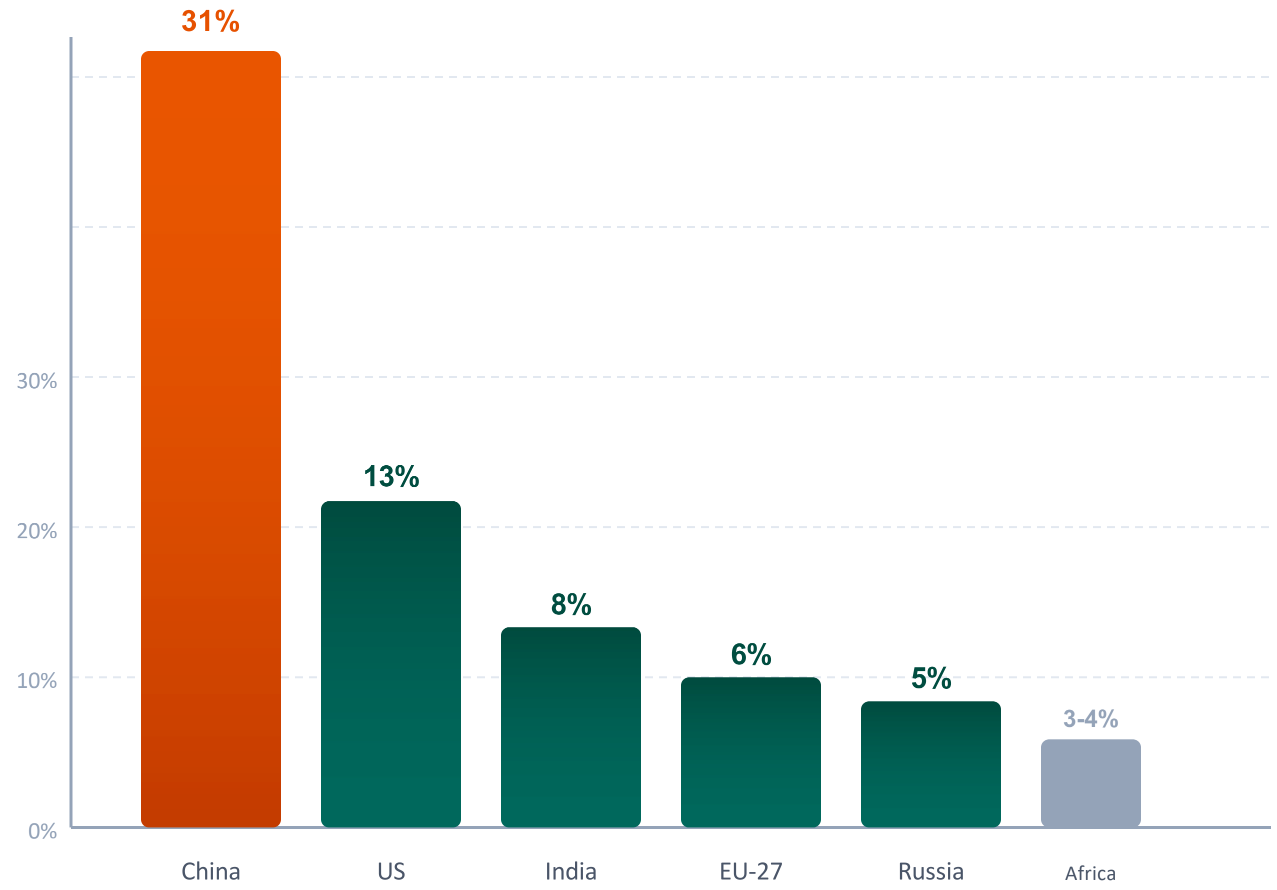
- Asia now produces ~half of global emissions, led by China
- Africa & South America each only 3–4% — comparable to international aviation & shipping

Historical responsibility is dominated by Europe and the US — a key factor in climate justice debates

Source: Global Carbon Budget 2025 via Our World in Data

China Now Emits More Than One-Quarter of Global CO_2

China's share (~31%) exceeds the combined emissions of the next four largest emitters



Country/Region Breakdown (2024)

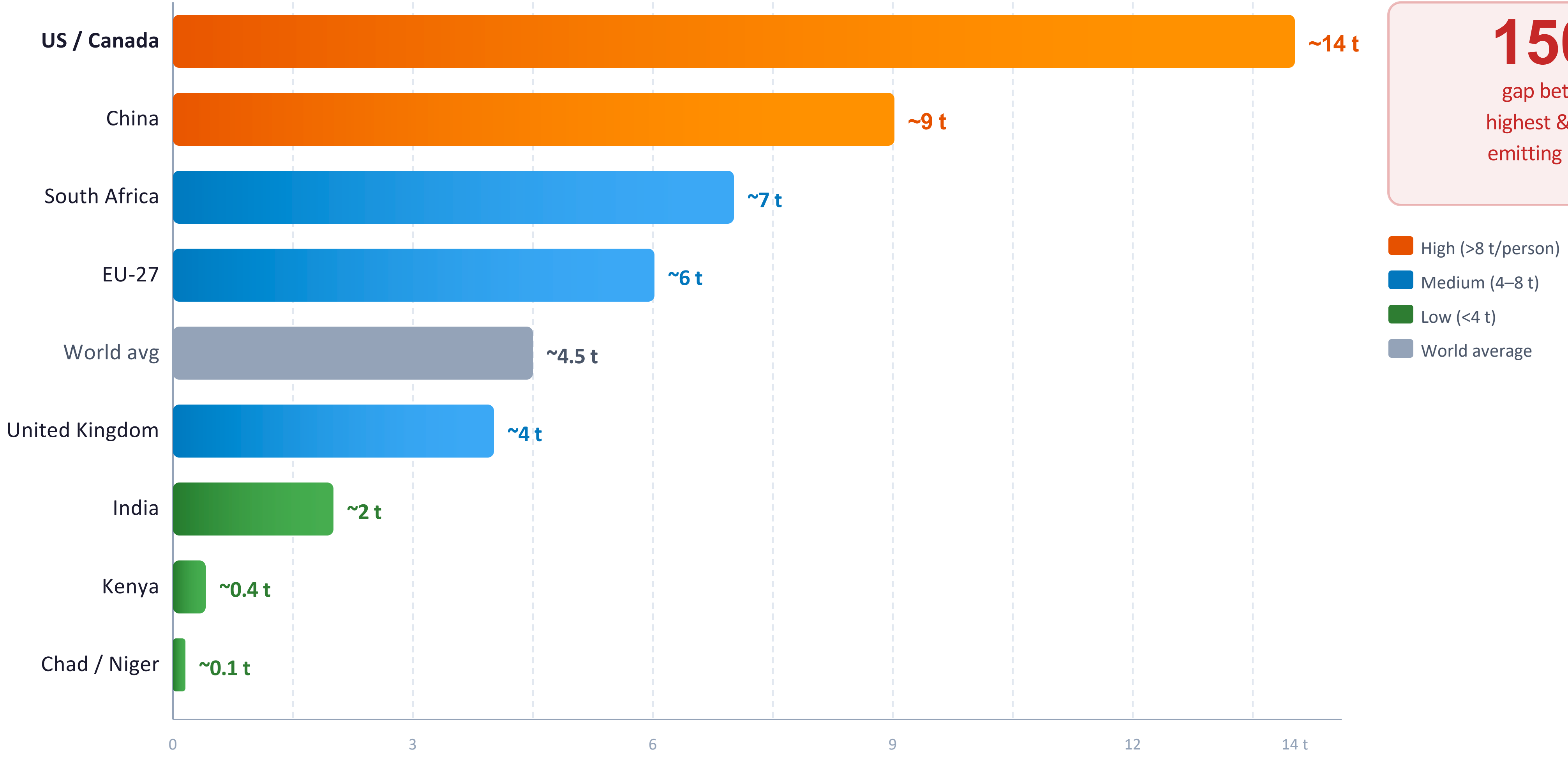
Country / Region	Share	Trend
China	~31%	Rising*
United States	~13%	Declining
India	~8%	Rising
European Union (27)	~6%	Declining
Russia	~5%	Stable
Africa (total)	~3–4%	Slow rise
South America (total)	~3–4%	Stable
Int'l aviation & shipping	~3%	Rising

*China's growth rate is slowing but absolute level still rising

Source: Global Carbon Budget 2025 via Our World in Data

Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada

A 150x gap: the average American emits in under 2 days what a person in Niger emits in a year

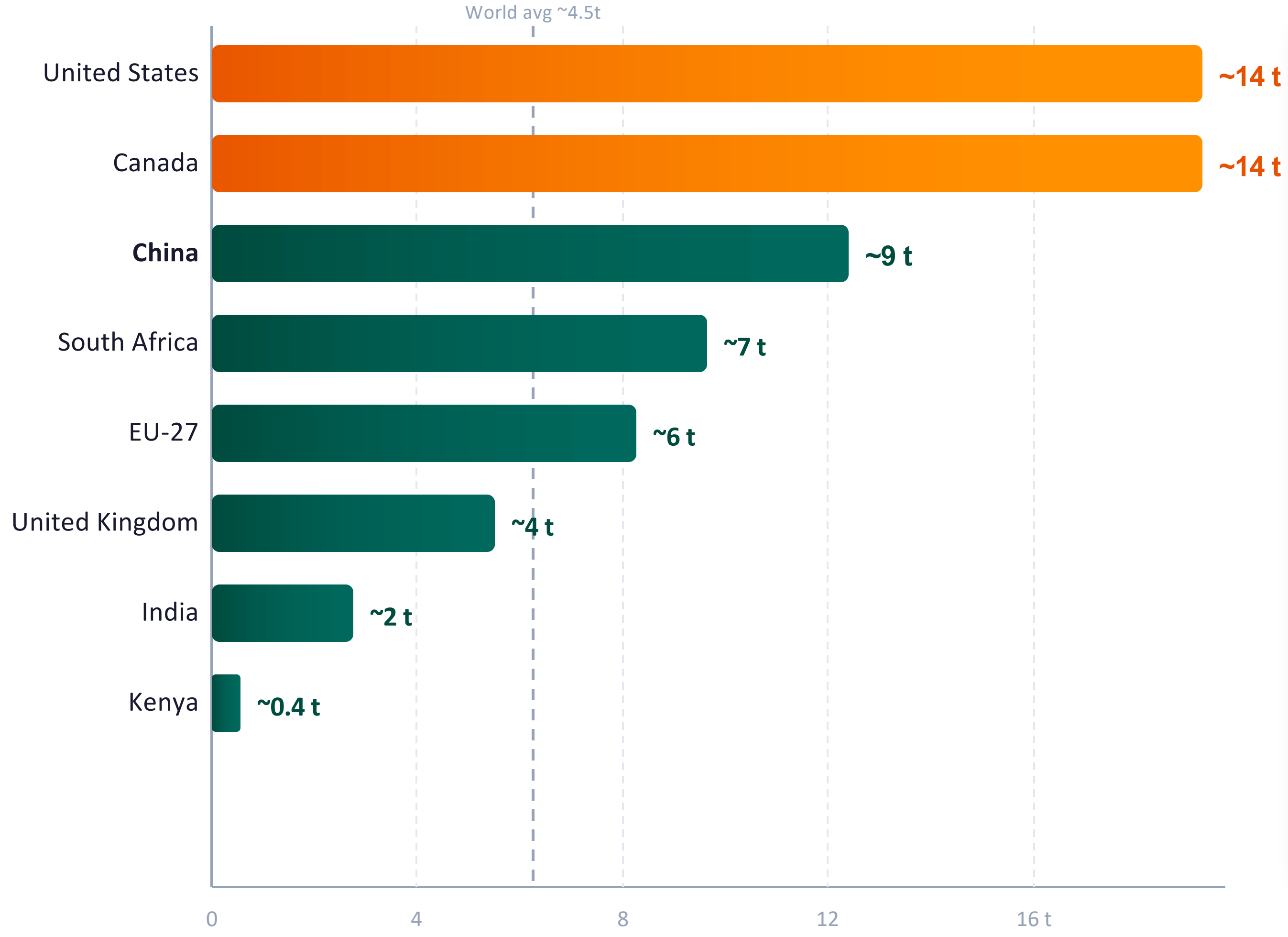


150x
gap between
highest & lowest
emitting nations

Source: Global Carbon Budget 2025 via Our World in Data

Per Capita Emissions Vary Widely:
~14 t in the US vs. ~9 t in China and ~2 t in India

China's per capita emissions remain below the US despite being the largest total emitter



Key Comparisons

US per capita
~14 t

vs

China per capita
~9 t

- US per capita is 1.5x China's
- China: largest total emitter but still below US per capita
- India remains very low at ~2 t, far below world average
- UK and France emit far less per capita than the US despite similar living standards

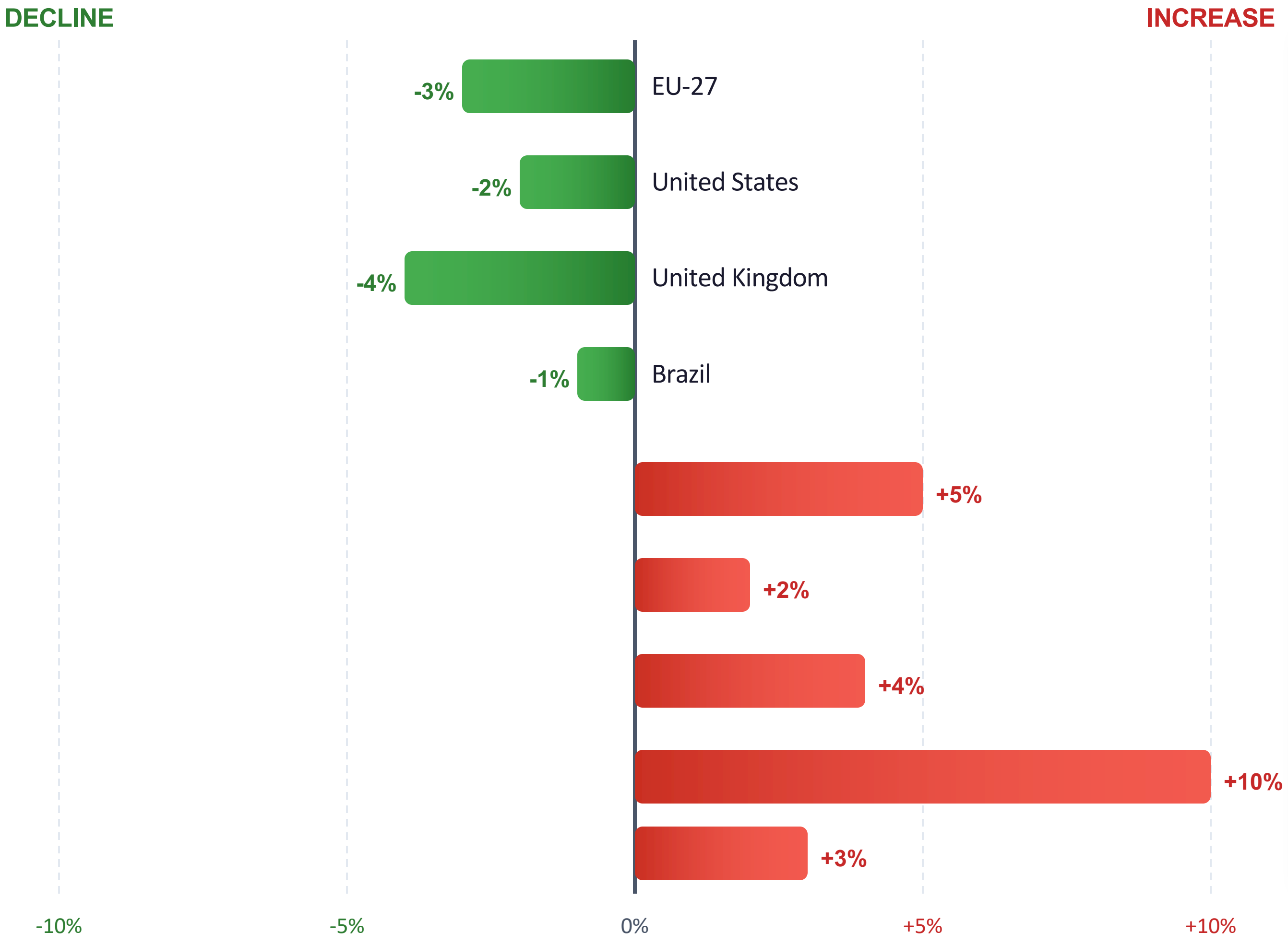
China's ~9 t/person vs US ~14 t/person: the gap in individual responsibility is narrower than total emission gap

Source: Global Carbon Budget 2025 via Our World in Data

2024 Snapshot: Many Western Nations Cut Emissions

While Parts of Asia and Africa Grew

The 2024 picture is mixed: developed nations are decarbonizing, but growth elsewhere offsets some gains



Emissions Declining

- US, EU, and UK saw **continued emission declines in 2024**
- Structural shift: clean energy replacing fossil fuels in the West

Emissions Rising

- Russia and parts of Southeast Asia saw increases
- Central Asia: increases **exceeding 10% in some nations**
- Several African nations also saw rising emissions

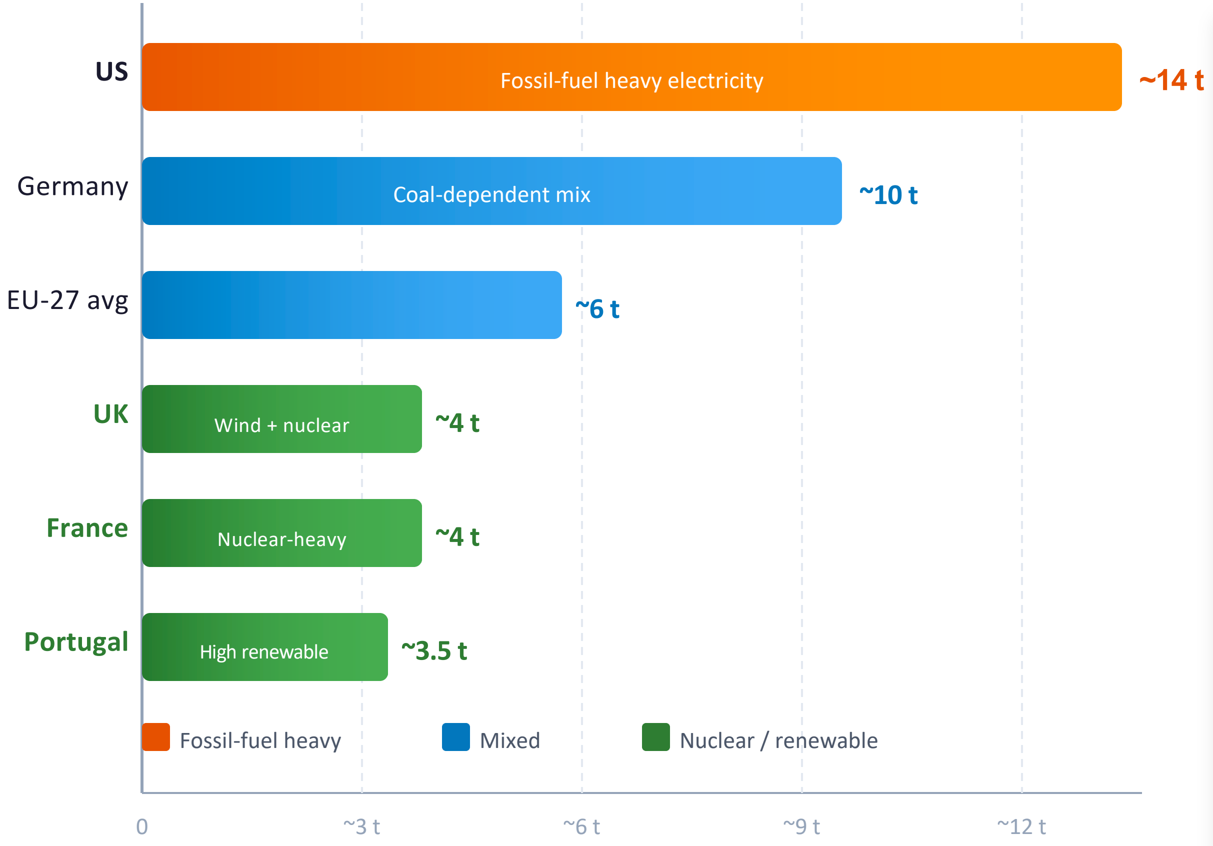
Increases in Asia and Africa partially offset Western declines

Source: Global Carbon Budget 2025 via Our World in Data

Energy Mix Explains Why France and the UK Emit Far Less

Per Capita Than Germany or the US

Energy policy choices — not just wealth — determine per capita emissions



Energy Mix Determines Impact

Among wealthy nations with similar prosperity,
per capita emissions vary 3–4x due to energy mix

- US: high fossil-fuel share in electricity generation drives ~14 t per capita
- Germany: still relies on coal, higher than France despite similar GDP
- France: ~70% nuclear electricity keeps per capita at ~4 t
- UK: major offshore wind build-out has cut per capita below EU average

Energy policy and technology choices can decouple prosperity from emissions — a crucial lesson for developing nations

Source: Global Carbon Budget 2025 via Our World in Data

Key Takeaways: Responsibility Is Shared but Unequal

No Peak Yet

Global CO_2 emissions from fossil fuels exceed **35 billion tonnes** per year and have not yet peaked, despite slowing growth.

The window for 1.5°C is closing.

China: Largest but Not Richest

China is the largest annual emitter at **~31% of global CO_2**, but per capita emissions (~9 t) still remain below the US (~14 t).

Total vs. per capita: different stories.

Historical Responsibility

Europe and the US dominated emissions historically **(>90% in 1900)** but now account for less than one-third.

Cumulative emissions matter for justice.

150x Per Capita Gap

A 150x gap persists between the richest emitters **(US/Canada/Australia ~14–15 t)** and the poorest nations (~0.1 t in Sub-Saharan Africa).

Inequality shapes climate policy and equity.

Policy Can Decouple

Energy policy and technology choices can **decouple prosperity from emissions.** France and the UK emit far less per capita than Germany or the US.

Hope: different choices yield different outcomes.

Shared but unequal responsibility demands both aggregate reduction and per capita equity

Source: Global Carbon Budget 2025 via Our World in Data